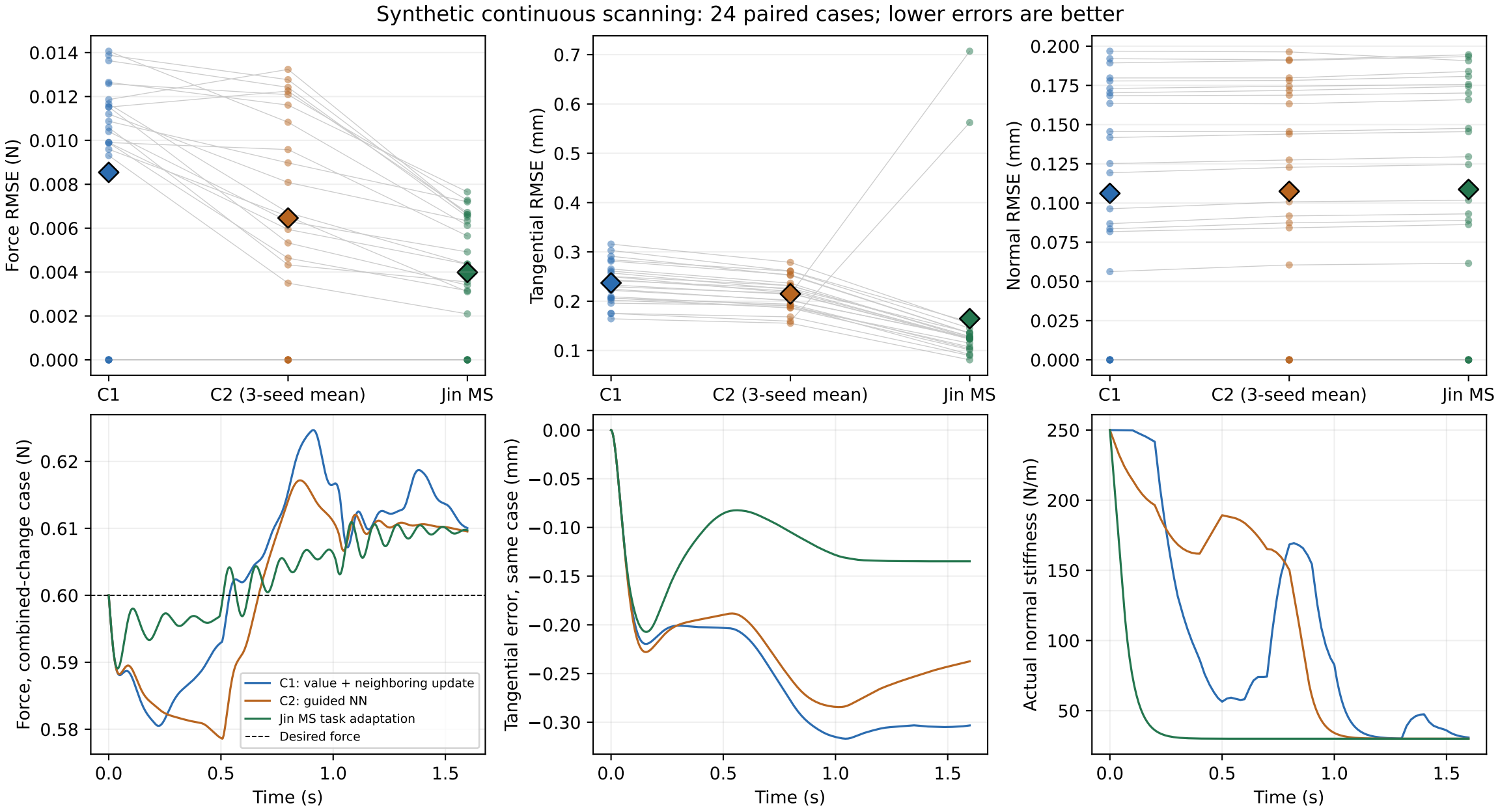
Synthetic continuous scanning: 24 paired cases; lower errors are better
Force RMSE (N)
Tangential RMSE (mm)
Normal RMSE (mm)
C1
C2 (3-seed mean)
Jin MS
Force, combined-change case (N)
Tangential error, same case (mm)
Actual normal stiffness (N/m)
Time (s)
C1: value + neighboring update
C2: guided NN
Jin MS task adaptation
Desired force